

Import Packages

```
In [4]: import numpy as np
import pandas as pd
```

1. Array (Numpy)

1-1. Creation of Array Data

1-1-1. Define Array Data by Assigning Values

- One-dimensional data (vector)

```
In [5]: array_1d = np.array([0,1,2])
array_1d
```

```
Out[5]: array([0, 1, 2])
```

"Numerical operations on array data are performed on each element of the array"

-> Very efficient in data analysis

```
In [6]: # List (basic data structures in Python)

a = [1,2,3]
b = [3,3,3]

a+b
```

```
Out[6]: [1, 2, 3, 3, 3, 3]
```

```
In [7]: # Array (numpy)

a = np.array([1,2,3]) # convert 'list' data into 'array' data
b = np.array([3,3,3])

a+b
```

```
Out[7]: array([4, 5, 6])
```

- Two-dimensional data (matrix)

```
In [8]: array_2d = np.array([[10,11,12],[20,21,22]])
array_2d
```

```
Out[8]: array([[10, 11, 12],
               [20, 21, 22]])
```

- Three-dimensional data

```
In [9]: array_3d = np.array([ [10,11,12],[20,21,22] , [[100,111,112],[120,121,122]] ])
array_3d
```

```
Out[9]: array([[[ 10,  11,  12],
                 [ 20,  21,  22]],

               [[100, 111, 112],
                 [120, 121, 122]]])
```

1-1-2. Generate Array Data Filled with 0 or 1

```
In [10]: ary_01 = np.zeros(5)
         ary_01
```

```
Out[10]: array([0., 0., 0., 0., 0.])
```

```
In [11]: ary_02 = np.zeros((2,3))
         ary_02
```

```
Out[11]: array([[0., 0., 0.],
                [0., 0., 0.]])
```

```
In [12]: ary_11 = np.ones(5)
         ary_11
```

```
Out[12]: array([1., 1., 1., 1., 1.])
```

```
In [13]: ary_12 = np.ones((2,3))
         ary_12
```

```
Out[13]: array([[1., 1., 1.],
                [1., 1., 1.]])
```

1-1-3. Generate Array Data by Designating a Range

`np.arange(a)` : Generate an array greater than or equal to 0 and less than 'a'

```
In [14]: ary_range_1 = np.arange(5)
         ary_range_1
```

```
Out[14]: array([0, 1, 2, 3, 4])
```

`np.arange(a,b)` : Generate an array greater than or equal to 'a' and less than 'b'

```
In [15]: ary_range_2 = np.arange(1,10)
         ary_range_2
```

```
Out[15]: array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

`np.arange(a,b,c)` : Generate an array greater than or equal to 'a' and less than 'b' with an interval 'c'

```
In [16]: ary_range_3 = np.arange(1,10,0.5)
         ary_range_3
```

```
Out[16]: array([1. , 1.5, 2. , 2.5, 3. , 3.5, 4. , 4.5, 5. , 5.5, 6. , 6.5, 7. ,
                7.5, 8. , 8.5, 9. , 9.5])
```

Range Definition of Basic Python

`range(a)` : Generate a range greater than or equal to 0 and less than 'a'

- It is not an array data. (mainly applying it for loops)

```
In [17]: range_1 = range(5)
         range_1
```

```
Out[17]: range(0, 5)
```

`range(a,b)` : Generate a range greater than or equal to 'a' and less than 'b'

```
In [18]: range_2 = range(1,10)
         range_2
```

```
Out[18]: range(1, 10)
```

1-2. Array Data Indexing

1-2-1. Extract a value from a specific position

```
In [19]: ary_1d = np.arange(1,10)
         ary_1d
```

```
Out[19]: array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [20]: ary_1d[0]
```

```
Out[20]: np.int64(1)
```

```
In [21]: ary_2d = np.array([[1,2,3] , [4,5,6] , [7,8,9]])
         ary_2d
```

```
Out[21]: array([[1, 2, 3],
               [4, 5, 6],
               [7, 8, 9]])
```

```
In [22]: ary_2d[0][0]
```

```
Out[22]: np.int64(1)
```

1-2-2. Extract values from specific range

```
In [23]: ary_1d[:]
```

```
Out[23]: array([1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
In [24]: ary_2d[1][0:2]
```

```
Out[24]: array([4, 5])
```

2. Data Frame (Pandas)

2-1. Creation of Data Frame

- Generally, DataFrame is defined by converting other data format

2-1-1. Define a DataFrame by Assigning Values

```
In [25]: df_1 = pd.DataFrame([11,12,13])
         df_1
```

```
Out[25]:
```

	0
0	11
1	12
2	13

```
In [26]: df_2 = pd.DataFrame({'A':['a','b','c'] , "B":[21,22,23] , "C":[31,32,33]})
         df_2
```

```
Out[26]:
```

	A	B	C
0	a	21	31
1	b	22	32
2	c	23	33

2-1-2. Generate a DataFrame by Converting an Array Data

```
In [27]: ary_2d
```

```
Out[27]: array([[1, 2, 3],  
               [4, 5, 6],  
               [7, 8, 9]])
```

```
In [28]: df_3 = pd.DataFrame(ary_2d)  
df_3
```

```
Out[28]:
```

	0	1	2
0	1	2	3
1	4	5	6
2	7	8	9

2-2. Data Frame Indexing

```
In [29]: df_3.iloc[:2, 0:3]
```

```
Out[29]:
```

	0	1	2
0	1	2	3
1	4	5	6

```
In [ ]:
```